

h34 2023

Q3. (b).

$$O = \begin{bmatrix} I \\ H\Phi \\ H\Phi^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0.2 & 0 \\ 1 & 0.4 & 0.2 \end{bmatrix}$$

has full row rank.
observable.